SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Industry Problem Solving Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO2: R Programming Structures, Different types of loops such as: for (), while() loops, Control Statements, Looping Over Non-Vector Sets, - If-Else, Arithmetic and Boolean Operators and values, Default Values for Argument, Return Values, Deciding Whether to explicitly call return- Returning Complex Objects, Functions are Objective, No Pointers in R, Recursion. (BL4-BL5)

Assessment Tool Weightage: 25%

Code of Conduct:

*I **A. Midhuna (192521302)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

A. Midhuna
Signature of the student:

QUESTIONS: **Total Marks: 100**

Scenario

A retail company operates stores across multiple cities. The management has collected customer transaction data containing:

- Customer_ ID
- Gender
- Age
- City
- Product_ Category
- Purchase _Amount
- Monthly_ Income
- Customer_ Rating
- Membership_ Status
- Store_ Type

The company wants to improve customer satisfaction, optimize marketing strategies, and increase sales using data-driven decisions.

Question 1

The management wants to understand the overall spending behavior of customers. Calculate and interpret appropriate measures of central tendency (mean, median, and mode) for Purchase Amount. Based on the results, recommend which measure best represents customer spending patterns.

ANSWER:

Coding:

```
# Purchase Amount data

purchase <- c (500, 700, 700, 800, 1000, 1200, 1500)

# Mean

mean_ value <- mean(purchase)
```

Median

median_ value <- median(purchase)

Mode

mode_value <- as. numeric (names(which.max(table(purchase))))

cat ("Mean:", mean_value, "\n")

cat ("Median:", median_value, "\n")

cat ("Mode:", mode_value, "\n")

Output:



Question 2

Analyze the variability in customer spending using measures of dispersion such as range, variance, standard deviation, and interquartile range. Identify whether spending behavior is consistent across customers and justify your findings.

ANSWER:

Coding:

purchase <- c (100, 200, 200, 300, 400, 500, 600)

cat ("Range =", diff(range(purchase)), "\n")

cat ("Variance =", var(purchase), "\n")

cat ("Standard Deviation =", sd(purchase), "\n")

```
cat ("Interquartile Range =", IQR (purchase), "\n")

if (sd(purchase) < 200) {

  cat ("Spending is consistent")

} else {

  Cat ("Spending varies across customers")

}
```

Output:



Question 3

The marketing team suspects that customer ratings differ significantly between premium and regular members. Perform an exploratory analysis of Customer Rating and summarize the key patterns using suitable statistical measures and visualizations.

ANSWER:

Coding:

```
rating <- c (4, 5, 3, 4, 5, 2, 3, 4)

membership <- c ("Premium", "Premium", "Premium", "Premium",

                 "Regular", "Regular", "Regular", "Regular")
```

premium <- rating [membership == "Premium"]

regular <- rating [membership == "Regular"]

cat ("Premium Mean =", mean(premium), "\n")

cat ("Regular Mean =", mean(regular), "\n")

cat ("Premium Median =", median(premium), "\n")

cat ("Regular Median =", median(regular), "\n")

cat ("Premium Minimum =", min(premium), "\n")

cat ("Premium Maximum =", max(premium), "\n")

cat ("Regular Minimum =", min(regular), "\n")

cat ("Regular Maximum =", max(regular), "\n")

Output:

```
rating <- c(4, 5, 3, 4, 5, 2, 3, 4)
membership <- c("Premium", "Premium", "Premium", "Premium",
                "Regular", "Regular", "Regular", "Regular")

premium <- rating[membership == "Premium"]
regular <- rating[membership == "Regular"]

cat("Premium Mean =", mean(premium), "\n")
cat("Regular Mean =", mean(regular), "\n")
cat("Premium Median =", median(premium), "\n")
cat("Regular Median =", median(regular), "\n")

cat("Premium Minimum =", min(premium), "\n")
cat("Premium Maximum =", max(premium), "\n")
cat("Regular Minimum =", min(regular), "\n")
cat("Regular Maximum =", max(regular), "\n")
```

```
Premium Mean = 4
Regular Mean = 3.5
Premium Median = 4
Regular Median = 3.5
Premium Minimum = 3
Premium Maximum = 5
Regular Minimum = 2
Regular Maximum = 5
```

Question 4

Investigate the relationship between Monthly Income and Purchase Amount using bivariate analysis techniques. Interpret the strength and direction of the relationship and discuss its implications for targeted marketing campaigns.

ANSWER:

Coding:

income <- c (20000, 30000, 40000, 50000, 60000, 70000)

purchase <- c (200, 300, 350, 500, 600, 700)

cat ("Correlation =", cor (income, purchase), "\n")

cat ("Mean Income =", mean(income), "\n")

cat ("Mean Purchase =", mean(purchase), "\n")

model <- lm (purchase ~ income)

cat ("Predicted Purchase =", predict (model,

data. Frame (income = 80000)), "\n")

cat ("Conclusion: Strong positive relationship\n")

cat ("Use premium products for high-income customers.")

Output:

```
income <- c(20000, 30000, 40000, 50000, 60000, 70000)
purchase <- c(200, 300, 350, 500, 600, 700)

cat("Correlation =", cor(income, purchase), "\n")
cat("Mean Income =", mean(income), "\n")
cat("Mean Purchase =", mean(purchase), "\n")

model <- lm(purchase ~ income)
cat("Predicted Purchase =", predict(model,
    data.frame(income = 80000)), "\n")

cat("Conclusion: Strong positive relationship\n")
cat("Use premium products for high-income customers.")
```

```
Correlation = 0.9943614
Mean Income = 45000
Mean Purchase = 441.6667
Predicted Purchase = 796.6667
Conclusion: Strong positive relationship
Use premium products for high-income customers.
```

Question 5

The company wants to determine whether male and female customers differ in their average purchase amount. Formulate appropriate hypotheses, perform a T-test, and interpret the results to support business decision-making.

ANSWER:

Coding:

```
male <- c (500, 600, 450, 550)

female <- c (400, 450, 500, 350)

cat ("Male Mean =", mean(male), "\n")

cat ("Female Mean =", mean(female), "\n")

p <- t.test (male, female) $p.value

cat ("P-value =", round (p, 3), "\n")

if (p < 0.05) {

  cat ("Significant difference")

} else {

  Cat ("No significant difference")

}
```

Output:

```
male <- c(500, 600, 450, 550)
female <- c(400, 450, 500, 350)

cat("Male Mean =", mean(male), "\n")
cat("Female Mean =", mean(female), "\n")

p <- t.test(male, female)$p.value
cat("P-value =", round(p, 3), "\n")

if (p < 0.05) {
  cat("Significant difference")
} else {
  cat("No significant difference")
}
```

```
Male Mean = 525
Female Mean = 425
P-value = 0.071
No significant difference
```

RUBRICS FOR CALCULATION

Industry Problem Solving Task					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Application of Techniques	25	Correctly applies central tendency, dispersion, and bivariate analysis techniques.	Applies most EDA techniques correctly with minor omissions.	Applies basic EDA techniques with limited accuracy.	Demonstrates limited or incorrect application of EDA techniques.
Statistical Analysis & Testing	25	Correctly performs and interprets T-test, ANOVA, and Chi-square analysis.	Performs statistical tests with minor interpretation errors.	Performs basic statistical analysis with limited interpretation.	Demonstrates poor understanding of statistical tests.
Problem Analysis & Data Interpretation	20	Accurately analyzes data patterns and derives meaningful insights.	Provides reasonable analysis with adequate interpretation.	Provides limited analysis and interpretation.	Analysis is weak or incorrect.
Decision-Making & Business Recommendations	20	Provides data-driven recommendations that effectively address the business problem.	Recommendations are relevant and supported by analysis.	Recommendations are basic with limited justification.	Recommendations are weak, unsupported, or missing.
Organization & Completeness	10	Response is well-structured, complete, and addresses all requirements.	Mostly complete and organized.	Partially complete with some missing components.	Incomplete and poorly organized.